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SURVEY PAPER

New trends in didactic research in university mathematics education

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Abstract
Recent research in university mathematics education has moved beyond the traditional focus on the transition from secondary to tertiary education and students' understanding of introductory courses such as pre-calculus and calculus. There is growing interest in the challenges students face as they move into more advanced mathematics courses that require a shift toward formal reasoning, proof, modeling, and problem-solving skills. This survey paper explores emerging trends and innovations in the field, focusing on three key areas: innovations in teaching and learning advanced mathematical topics, transitions between different levels and contexts of mathematics education, and the role of proof and proving in advanced university mathematics. The survey reflects the evolving landscape of mathematics education research and addresses the theoretical and practical challenges of teaching and learning advanced mathematics across various contexts.

1 Introduction

For many years, research in university mathematics education was devoted to the transition from secondary to tertiary education and to studying students' understanding of introductory courses such as pre-calculus, calculus, and introduction to linear algebra. Only some research has focused on more advanced topics such as analysis or abstract algebra.

Recently, there has been a strong movement in university mathematics education to learn more about the learning and teaching of more advanced mathematics courses in which proof and proving are central themes. In addition, another critical, difficult transition for students has become more apparent at the university level: from learning in introductory courses to learning in advanced courses. This transition involves a greater emphasis on formal reasoning, proof, modeling, and problem-solving skills, requiring more effort to prepare students for such a transition.

The papers in this special issue focus on one of these topics and represent new trends or innovations in research in university mathematics education. The novelty may relate to the research topic or an innovative theoretical or methodological approach.

We structure our papers around these three themes: innovation, transition, and proof and proving.

Each of these three sections presents an overview of the respective theme at the advanced university level, including the papers in this special issue, with some papers appearing in more than one section. This allows us to provide a structured view of the state of the art of new trends in advanced university mathematics education as they appear in recent books, journal special issues, and conference proceedings. It also allows us to highlight new cross-cutting trends and related work in different topic areas.

2 Innovation in university mathematics education

What is innovation? The definition in the Oxford dictionary can be interpreted as doing something in a new way if we consider innovation in education. From Evans and Leppmann's (1970) perspective, it has two components: a novel idea or item and the change resulting from its adoption. These aspects are related to the idea that when an innovation is introduced in an educational context, it results in better learning of that specific content by students than by students following traditional teaching (Serdyukov, 2017). In university mathematics, it can be added that innovation should

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be based on research in mathematics education. It should help students learn particular mathematical content better than traditional teaching (Florensa et al., 2022). Interactive instruction can involve changing students' roles in activities and practices in the classroom (Beauchamp & Kennewell, 2010).

In their survey about innovative approaches to teaching mathematics in higher education, Abdulwahed et al. (2012) found that university mathematics has been slow to adopt innovations. Teaching at this level mostly continues to use lectures to introduce theory with some examples as the primary communication tool without paying much attention to students. They also mention the surge of calls for innovation in pedagogical approaches, mainly based on constructivist approaches. That is, considering students' previous knowledge, contextualized problems, collaborative work, and teachers' interaction to guide students and use formative assessments (Abdulwahed et al., 2012).

Although most innovations seem to be mainly concerned with elementary and secondary classroom practices, innovative research in advanced mathematics education has been and is today concerned with research-based teachers' development, assessment of proposed practices, participation of mathematicians in mathematics education research, teacher-student interaction, and the role technology plays in fostering students learning. As society's problems evolve, research in advanced mathematics education is facing new challenges. New innovative approaches have appeared or are needed in response.

Innovation in research on university mathematics so far can be related to different aspects. We thus start discussing innovations in the development of theoretical perspectives, followed by innovation in teaching, including the use of technology and innovation in motivation and values on engagement and learning experiences, as suggested by the papers related to this topic in this issue.

2.1 Innovation in the development of theoretical perspectives

Research in mathematics education is characterized by many theoretical perspectives based on different epistemological approaches. Efforts have been made to compare and network different theoretical positions to look for similarities and differences between the theories and thus decide whether it is possible to find a way in which they can safely complement each other and support the design of innovations (Bikner-Ahsbahs & Prediger, 2014; Nardi et al., 2014). In what follows, we focus on examples of such perspectives in recent university mathematics education.

Some theories are based on theoretical foundations and research and didactic methodologies. They can be used coherently in research and practice. Other emerging theoretical approaches are sometimes limited in application but appropriate for more specific teaching intervention studies. In view of the multiplicity of theories, projects have been launched to compare and sometimes to integrate different theories (Bikner-Ahsbahs & Prediger, 2014). The results of these efforts are varied. Sometimes, integrating theories enables researchers to conduct a detailed analysis of a specific didactic phenomenon. Others give rise to what can be considered a new theory.

This was the case in a study in which the theory of semiotic representations (Duval, 2006) and Action, Process, Object, Schema (APOS) theory were partially integrated (Trigueros & Martinez-Planell, 2010). Another possibility is to compare different theories to facilitate their leverage (e.g., Jaworski & Robert, 2018) or their integration in a productive manner (Bikner-Ahsbahs & Prediger, 2014). It can even be possible to develop a new framework resulting from the dialogue between theories (Bosch et al., 2017). In doing so, researchers can better account for the multifaceted nature of educational phenomena and obtain a holistic understanding of the available data while fostering collaboration across theoretical boundaries. The collaborative work of researchers in the International Network for Didactic Research in University Mathematics (INDRUM) conferences, the publication Research and Development of University Mathematics Education (Durand-Guerrier et al., 2021), and the book resulting from European collaboration in the Platinum Project (Gómez-Chacón et al., 2021) are examples of the potential of collaborative projects.

Two papers included in this issue exemplify this category: Lankeit and Biehler (2024, this issue), and Villabona Millán et al., (2024, this issue). Lankeit and Biehler (2024) developed an innovative theoretical model of meaning for advanced mathematical concepts to be used as a framework to analyze advanced mathematical textbooks. Their work builds on Hußmann and Prediger's (2016) notion of formal and semantic levels and introduces the concept of facets. They categorize meaning facets across various contexts of interpretation. Following Richenhagen (1985), they differentiate definition variants for mathematical objects into constructive and relational-descriptive types, forming a basis for the analysis of meaning facets which can be categorized across various contexts of interpretation. The authors use their model to analyze the meaning facets of derivative and differentiability of multivariable functions.

Villabona Millán et al., (2024, this issue) present an innovative study on understanding mathematical infinity. They analyze how mathematical infinities are constructed, leading them to reconsider APOS theory based on experimental findings from graduate students. Their research highlights that individuals can perform some *actions* related to infinity in certain contexts but not in others (Dubinsky et al., 2013; Villabona Millán et al., 2022). Their experimental study

utilizes an infinite union of finite subsets as context. They identified different types of Actions linked to the construction of what they term "facets" of the *object* acted upon. This new construct prompts a reconsideration of the nature of infinite objects and suggest didactic approaches to enhance understanding of mathematical infinity.

2.2 Innovations related to teaching practices and the use of technology

The traditional lecture remains the primary teaching method for mathematics in higher education. New research theory-based teaching approaches have been developed by the research community in advanced mathematics education to foster active roles for both teachers and students in the university classroom. However, research into teachers' practices and their communication with students has progressed slowly.

Research in university mathematics highlights that the mathematical concepts to be taught are often challenging for students. Innovative teaching and learning strategies aim to prepare students to effectively apply their mathematical knowledge in practice and when teaching in schools. Significant learning of abstract mathematics concepts depends on teachers' deep understanding of mathematics and their ability to clearly communicate complex ideas. Continuous professional development is crucial for equipping teachers with the skills needed to support students in advanced mathematics. Furthermore, teachers' comprehension of curriculum structure and content, along with assessment methods, shapes students' learning experiences. Balancing rigor and accessibility, alongside effective teaching practices, is vital for providing students with opportunities for meaningful learning. The complexity of teaching and learning in advanced mathematics courses calls for further research to explore effective strategies for supporting teachers and students in this challenging educational context (Winsløw et al., 2021).

Florensa et al. (2022) have investigated innovative teaching strategies designed to enhance learning in advanced mathematics education, focusing on the challenges of teaching complex mathematical concepts through student-centered approaches. Research on interactive instruction suggests that it can change students' roles in classroom activities (Beauchamp & Kennewell, 2010).

Research-based teaching strategies in advanced mathematics education prioritize enhancing students' learning experiences. Many proposals utilize active learning techniques, such as problem-based learning (PBL) and collaborative group work, encouraging students to engage directly with mathematical problems and discuss viable solutions. PBL aims to promote deep reflection to help students understand and retain complex concepts (Florensa et al., 2022; Nardi & Knuth, 2017). Inquiry-based learning (IBL) (Artigue & Blomhøj, 2013; Rasmussen & Kwon, 2007) is a key strategy for fostering critical thinking and problem-solving skills. IBL encourages students to take ownership of their learning by posing questions, exploring various approaches, and developing their solutions to mathematical problems. The teacher's role is to support students with feedback and guidance, adapting to class size. The characteristics of IBL depend on the theoretical framework used for designing activities and the teacher's approach during classes. Recently, this methodology has been analyzed in the Platinum project where various approaches and innovative teaching experiences are discussed (Biehler et al., 2022; Gómez-Chacón et al., 2021).

Among the contributions received for this issue, the papers by Soto et al., (2024, this issue), Hanke (2024, this issue), and Trigueros et al., (2024, this issue) relate to the two first parts of our discussion on innovation in learning and teaching advanced mathematics.

Soto et al., (2024, this issue) present a detailed analysis of how a mathematician taught the Cauchy-Riemann (CR) equations during three class sessions. They use embodiment as a theoretical framework and contribute to it by rethinking the sense of "body in and of mathematics" used by de Freitas and Sinclair (2014). Through videotape analysis, the authors search for the "virtual dimension" (Châtelet, 1993), that is, the visible and non-visible agents binding and structuring the didactic practice in undergraduate mathematics. They use videotapes to focus on the teaching of a mathematician who used embodiment to animate and bind formal aspects of the CR equations. The data analysis results can help authors find out how the mathematician achieves it and illustrate the virtual dimension as they relate to the CR equations. The results show that the mathematician engaged with mathematical objects originally and creatively. The researchers found the use of metaphors and dynamism related to the CR equations' virtuality in their data. The mathematicians' explanations were also creative and dynamic, assembling creative acts using algebraic and geometric inscriptions. He tended a bridge between the virtual and the abstract. The authors discuss the possible didactic implications that may help teachers to stress, for example, the mobility of conformal maps. The authors suggest some didactic practices for teachers to include in their classes to support the binding of mathematical concepts. The innovation of this study lies in introducing an original approach that is new to university mathematics education.

Hanke (2024, this issue) explores the possibility of relating interpretations for real integrals to those in complex analysis by asking experts about their interpretation of complex path integrals and analysing their responses. He uses the commognitive framework (Sfard, 2008) and introduces the notion of intuitive mathematical discourse to explore

experts' interpretations of complex path integrals, referred to as discursive images. The analysis shows that even experts have difficulties interpreting complex path integrals, but provided evidence of the various aspects of this topic. The author identifies the discursive frames that govern the production of discourse images. It is shown that experts develop multiple interpretations of complex path integration in relation to the identified frames.

Trigueros et al., (2024, this issue) propose an innovative didactic approach for introducing vector space in university linear algebra courses. They designed a didactic approach using APOS Theory focusing on the construction of prerequisites suggested in the literature, followed by work on sets with atypical elements and various binary operations for students to determine which sets are vector spaces. The study aimed to reflect on the possible benefits of the experience in terms of student learning. Students worked collaboratively in small groups, followed by whole class discussions with the teacher. Four students were interviewed at the end of the experience. Results showed that the approach allowed students to reflect on their knowledge and to recognise the need to prove statements using the appropriate approach to determine which sets were vector spaces.

Another innovative study on course design described in the transition section is the design of the geometry course for teachers by Hoffmann and Biehler (2024, this issue). Authors show that the introduction of interface work and tasks has a positive impact on the quality of mathematical clarification and student-oriented explanations.

One important innovation in advanced mathematics education is integrating technology into the classroom. Digital tools such as computer algebra systems (CAS), dynamic geometry software, and online platforms like MOOCs have revolutionized how students engage with mathematical concepts. These technologies allow for interactive and visual representations of abstract ideas, making complex topics more accessible. The appearance of new technologies has changed the way of learning and doing mathematics. The mathematical skills needed to calculate in the past have been replaced by others that require time and effort to master and require mathematical thinking. Students are more willing to use interactive resources they find online to practice and learn at their own pace (Sacristan et al., 2010). Students can also use online simulations to explore and understand mathematical concepts or use artificial intelligence (AI) to solve their doubts, to find responses easily, or in helping in proof-related tasks. Technology can be helpful, but students sometimes have difficulties understanding abstract concepts or misunderstand the mathematics these resources try to communicate. Studies on the use of these and other digital technologies and their limitations and concerns related to their use in the classroom are analyzed in two survey papers, namely Engelbrecht and Borba (2024) and Weigand et al. (2024).

Researchers underlined the role of programming in students' learning university mathematics since the 1990s (e.g., Dubinsky & Tall, 1991). Both researchers found that this practice had beneficial results in students' learning. These have recently been recognised, and several researchers have highlighted its role in students' understanding of advanced mathematical topics. (e.g., Buteau et al., 2019; Lockwood et al., 2019). Two innovative studies in this issue, Hanna et al., (2024, this issue) and Yoon et al., (2024, this issue), use AI technology to help students understand the meaning of proof and how to prove statements. These papers are discussed in the proof section.

2.3 Innovation in motivation and values on engagement and learning experiences

The role of motivation, beliefs, engagement, and learning experiences has received increased attention in elementary and secondary education. Studies exploring the emotional and affective aspects and analyzing their influence on students' experience with mathematics have highlighted the complex interplay between emotions, beliefs, and mathematics learning processes. Understanding the motivations and values that drive students to university courses or majors can help to create an environment that promotes engagement with abstract concepts, proofs, and possible applications (Goldin et al., 2014).

One innovative paper in this issue (Asada et al., 2024, this issue) explores studies results and their potential impact on student outcomes using the notions of intrinsic, utility, and attainment values (Eccles, 1984; Eccles & Wigfield, 2020)- Authors analyse the role of values in students' motivation in abstract algebra and the factors that lead them to place high intrinsic value on the subject. The results showed that all students considered intrinsic value as a guide to their motivation and engagement in the course, and that motivation guided their learning. The authors suggest that the theoretical and abstract nature of advanced mathematics increases the importance of intrinsic value for students, and that this was facilitated by collaborative work and the use of projects.

3 Transition

3.1 Transition in general

Gueudet et al. (2016) provide an overview of transitions in mathematics education. A basic distinction is made between epistemological and cognitive transitions on the one hand, and sociocultural transitions on the other. In the latter, they

distinguish different transition points related to university education:

1. "School to undergraduate courses,
2. Undergraduate courses to postgraduate courses,
3. University into employment, and
4. University into teaching." (p. 4)

These are transitions between institutions where different cultures of teaching and learning prevail and where different types of mathematics are involved. Such transitions have elements of continuity and discontinuity, the latter of which can create significant problems for the transitioning learner.

In the anthropological theory of didactics (ATD), the concept of institution has a broader meaning (Chevallard, 2019). From this perspective, each individual course as part of a university program can be considered as an institution itself, which can be characterized by a specific mathematical and didactic praxeology. Therefore, the transition between courses can also be seen as a transition between institutions. This is relevant, for example, between courses in mathematics for a specific audience, let us say engineering students, and courses in engineering, where mathematics embedded in an engineering context plays an important role, or courses in mathematics for teachers compared to courses in the didactics of mathematics for teachers. The transition from undergraduate to postgraduate courses can be thought of as a transition between educational institutions in the colloquial sense. However, it can also take place within the same institution, although the kind of mathematics is very different, if rigor and proof orientation play a much more fundamental role in postgraduate courses.

Gueudet et al. (2016) focus in more detail on three transitions: The double discontinuity between secondary school mathematics and university mathematics, the transition between in- and out-of-school mathematics, and the transition between educational institutions. The papers in this issue do not deal with the transition between in-school and out-of-school mathematics. However, there are similarities with the transition between mathematics in mathematics courses and courses in other disciplines or employment contexts.

3.2 The secondary to tertiary transition

Although none of the papers in this volume deal with the transition from secondary to tertiary education, we feel it is necessary to summarize important work on this transition as well, since the differences between secondary and tertiary mathematics education that have been analyzed are also relevant for the other partially inverse transitions, such as when future teachers "return" from tertiary education to secondary education. However, the measures to support the inverse transition must be very different.

The transition from secondary to tertiary education has been extensively studied in the last twenty years. Gueudet (2008) gave a kind of starting signal that triggered much more research and surveys, such as Thomas et al. (2015) or the papers in Chapter I of Rezat et al. (2014). Gueudet (2023) provides an update on the state of the art on the secondary-tertiary transition. This is part of a comprehensive special issue on the topic, to which Di Martino et al. (2023) contributed a review article with new research in the field.

The papers were analytical in the sense that they characterized the different types of mathematics, the different learning cultures in schools and universities, and the conditions for academic success, failure, or even dropout in relation to students' prior mathematical knowledge, affect, interest, identity, learning strategies, or learning behaviors. Some identified institutional conditions that were not favorable for the transition period. Sometimes, the consequences of redesigning the transition to improve the situation were studied. Hochmuth et al. (2022) have developed a theoretical approach for characterizing such transition support measures and have developed instruments for studying their effects. We also see a growing number of policy position papers from different academic groups, some of which see the solution to some of the problems in changing secondary and tertiary education, see DMV, GDM, & MNU (2019), for the debate in Germany. In the UK, many publications refer to this as the "mathematics problem", which dates back to the London Mathematical Society (1995).

Intervention studies are related to preparatory, bridging, and remedial courses, to support measures in the first year of university study, including the creation of dedicated student support centers, and to changes in the teaching–learning system. Some others are related to the lectures themselves (such as inquiry-based teaching) or to the accompanying tutorials, for example, better qualification of undergraduate and graduate teaching assistants (see references in the next section).

The transition from secondary to tertiary education has been one of the research foci of the khdm (German Center for University Mathematics Education) in Germany, starting with bridging courses (Bausch et al., 2014) to looking at the transition more generally (Hoppenbrock et al., 2016). A collection of innovative interventions from the khdm can be found in Biehler et al. (2021). The WiGeMath project (Hochmuth et al., 2022) at the khdm systematically and empirically analyzed support measures for the transition, such as bridging courses, student support centers (Matthews et al., 2012; Schürmann et al., 2020), and innovative lecture types in the first semester that directly address the transition; see also Liebendörfer et al. (2022) for the theoretical framework developed for a systematic structuring of support measures.

Similar comprehensive initiatives and research are documented by the Matric Center in Norway (https://www.uia.no/om-uia/senter-og-nettverk/matric/), the RUME community in the US (see the proceedings of the annual RUME conferences (http://sigmaa.maa.org/rume/Site/Proceedings.html), and the Sigma Network in the UK with its publication MSOR Connections (https://journals.gre.ac.uk/index.php/msor).

3.3 Klein's double discontinuity and the transition from university to the mathematics teachers' workplace

The transition from university to school teaching has received increasing attention in recent years. Wasserman et al. (2023) and the related special issue provide an overview of this topic, focusing on making university mathematics more relevant in secondary school teacher preparation. University studies also include the acquisition of scientific knowledge in the didactics of mathematics and pedagogical theories and principles of teaching. Often there is an integrated or subsequent internship or guided practice period before teachers can teach autonomously. The transition involves not only "knowledge" but also affect and identity (Gildehaus et al., 2024). There is a growing body of theoretical-analytical research on how to conceptualize the tertiary-secondary transition; see Scheiner and Bosch's (2023) comparison of Klein's approach, the didactic transposition approach (ATD), and Shulman's disciplinary transformation approach. It is important to understand the different praxeologies for mathematics at the secondary and tertiary levels from the perspective of the second discontinuity that limits the possibilities for transition.

There are very different praxeologies at the secondary and tertiary levels, as research at the secondary-tertiary level has shown. It needs to be clarified why, to what extent and in what sense knowledge from a different praxeology (here tertiary mathematics) is relevant at all for the different praxeology of secondary mathematics. The concrete relevance will depend on whether one wants to change secondary mathematics in the direction of tertiary mathematics, for example by giving more weight to conceptual rigor and proof, by emphasizing mathematical structures more than is currently the case (and as was aimed at in the Modern Math movement, see De Bock, 2023), or by integrating newer mathematical topics into secondary mathematics education, such as the mathematical background of artificial intelligence or difference equations and fractals.

It is often not emphasized enough that Klein himself saw the overcoming of the second discontinuity as relevant not only for the preparation of teachers for the curricula of the time, but also for supporting curricular change. He was part of a reform movement that emphasized a more visual-geometric, application-oriented approach with a focus on "functional thinking" (Weigand et al., 2019). In the modern mathematics reform movement, the transition from tertiary to secondary education was a combined effort of teaching teachers a new kind of mathematics and reforming the curricula at the same time so that the new teacher education was well suited to the new needs (De Bock, 2023).

Teacher education can encourage teachers in this direction. However, it may lead to frustration if the future conditions in secondary schools do not support such an innovation. Related to some innovative teaching experiments, Gueudet (2023) states:" *Will these students, once they become secondary school teachers, really implement such practices? This is very unlikely, due to the institutional constraints framing teaching practices"* (Gueudet, 2023, p. 171). Thus, the tertiary-secondary transition may require a deeper praxeological analysis to identify constraints and innovation opportunities. An example where the double discontinuity is analyzed from a praxeological (ATD) perspective is Winsløw and Grønbæk (2014).

Wasserman et al. (2023) emphasize that the evidence does not support the "trickle-down" hypothesis that university mathematics knowledge automatically translates into classroom practice. On the contrary, "Prospective secondary teachers have commonly criticized the lack of practical applicability of their university studies in mathematics to their future profession" (p. 720). However, there is considerable variation in practicing teachers' views of the relevance of university mathematics (Allmendinger et al., 2023; Zazkis & Leikin, 2010).

The high academic demands may be detrimental because they may contribute to high dropout rates among prospective teachers (Gueudet, 2008). This points to the challenge of subjectively convincing prospective teachers that their mathematical knowledge will be useful in the future. Hoffmann and Biehler (2023) call this the *subjective facet* of the second discontinuity. A relevant question, however, is not only how to make current university mathematics relevant to prospective teachers. It may also be the case that university mathematics courses need to be supplemented with mathematical topics that are relevant for teaching at the secondary level. A content example is Euclidean synthetic geometry, which is taught at the secondary level but often needs to find a place in programs for mathematics majors, which prospective teachers also attend. This is part of the *objective facet* of the second discontinuity. Another aspect brought forward is the notion of mathematical practices or habits of mind such as problem solving, representing, proving, defining, modeling, and generalizing (Wasserman et al., 2023, p. 725), which are considered relevant for secondary mathematics education but may not be sufficiently developed or remain implicit in university mathematics courses if they follow the "lecture paradigm".

Wasserman et al. (2023) review practical attempts at course redesign to bridge university mathematics and school mathematics by adding interface tasks or sections to mathematics courses or by developing specific capstone courses for teachers. These innovations use bottom-up and top-down approaches or both, depending on whether university mathematics content is motivated by school mathematics or applied to school mathematics. Other options are content-specific courses in the didactics of mathematics in parallel to mathematics courses that establish relations to school teaching (Hanke & Bikner-Ahsbahs, 2019; Hanke et al., 2021). For the development of mathematical practices, inquiry-based mathematics courses with a focus on teacher education are another option (Apkarian et al., 2023; Laursen & Rasmussen, 2019; Salazar Solórzano, 2015). The growing literature on inquiry-based university mathematics education (see, e.g., Biehler et al., 2022, and Gómez-Chacón et al., 2021) has not been fully exploited from a teacher education perspective.

Conceptualizing effects on the attitudes and knowledge of prospective teachers is a theoretical challenge, and empirical results of different conceptualizations are available for some of the interventions (see Wasserman et al., 2023 and the papers in the related ZDM special issue). How these innovative courses may also have long-term effects on the future teaching practice of preservice teachers is very difficult to study, requiring well-designed long-term longitudinal studies.

The following section will situate the three papers on this special issue in the developed framework. Hoffmann and Biehler (2024, this issue) focus on the same geometry course for secondary teachers as in Hoffmann and Biehler (2023), but this time emphasize symmetry, an advanced mathematical topic. The related course textbook (Hoffmann et al., 2024) differs from other geometry courses for secondary teachers, though details aren't provided here. Symmetry, crucial in secondary mathematics, is taught differently at university level (isometric mappings, symmetry group) compared to secondary school, where it is connected to practical activities. The study examines how prospective teachers' responses to a symmetry-related student problem improve after an interface week, showing growth in both mathematical understanding and explanatory ability. Their model can be applied to other tasks focusing on developing mathematical knowledge for teaching.

Grenier-Boley and Robert (2024, this issue) present a capstone course design that aims to overcome the second discontinuity with new principles. While no empirical study of an implementation is provided, the course design and its theoretical underpinnings are novel and may stimulate further research. The course, "Learning Arithmetic as a University Student, as a Teacher Would Want His Students to Learn," engages students in inquiry-based learning of arithmetic concepts and number theory. Students reflect on their learning processes and practice the design of inquiry-based school activities.

This design goes beyond simply adding interface weeks or inquiry-based tasks to include mathematical and pedagogical reflection and the design of learning activities for peers who represent future students. The concept of "double institutionalization", linking mathematical and didactic elements, is a valuable theoretical contribution.

A third paper of this special issue that can be related to transition is the paper by Durand-Guerrier (2024, this issue). It discusses transition issues associated with the real numbers in secondary and tertiary education. It fits nicely into previous research on this topic (Barquero & Winsløw, 2022; Branchetti & Durand-Guerrier, 2023; Durand-Guerrier, 2022; Winsløw & Grønbæk, 2014). The new insight, however, is related to how the completeness of R as different from the incompleteness of Q is relevant in contexts of proving the Intermediate Value Property and the Bolzano-Weierstrass theorem. Based on the epistemological analysis, a multi-proof activity is designed for prospective teachers to reflect on the role of the different completeness characterizations and the role of completeness in proving and reasoning at the tertiary and secondary levels.

3.4 Transitions between mathematics courses and courses in other university disciplines where mathematics is used

Similar to the secondary-tertiary transition, research is also devoted to better understanding these transitions inside the universities. Research is devoted to smoothing the discontinuities through intervention studies that may propose fully integrated approaches or approaches that add "interface" and application examples from subject X to a mathematics course for subject Y (see Schmidt & Winsløw, 2021; Wolf, 2017) or pay more explicit attention to the situated mathematical needs in subject courses (see Hennig et al., 2015, for electrical engineering).

Pepin et al. (2021) provide a recent review of innovative approaches in engineering education, which serves as an introduction to a special issue of IJRUME on this topic. Research on transitions and relationships between mathematics and other disciplines, with a focus on calculus, can be found in Dreyfus et al. (2023) and the special issue on calculus at the intersection of institutions, disciplines, and communities (see Biza et al., 2022, for its editorial). A trend is to use the Antropological Theory of the Didactic (ATD) to better understand different praxeologies and to design new learning trajectories for, e.g., engineering students (González-Martín et al., 2022; Peters, 2022).

A further focus from which Pepin et al (2021) analyzed the developments in teaching mathematics in and

for engineering is the *resource approach* to mathematics. The reviewed studies mainly focus on students' choice and combined use of resources (Pepin & Kock, 2021). Sabra and Tabchi (2024, this issue), however, explore teachers' use of resources, focusing on how their disciplinary backgrounds shape this process and its relation to transition issues. Drawing on ATD concepts like "knowledge to be taught" and "learned knowledge," they introduce the idea of resource connectivity, distinguishing between potential connectivity (in designed resources) and effective connectivity (in individual implementation). A key finding is the significant variability in how two teachers with different backgrounds implement the same co-designed resources. This approach, examining multiple teachers within the same institution, provides valuable insights into the variability between knowledge taught and knowledge to be taught. This focus on teacher variability is a valuable research interest that deserves much more attention in the future.

4 Proof and proving in advanced university mathematics

4.1 Introduction

While research in university mathematics education on proof and proving has been developing for decades, new research directions are now emerging. In general, proof and proving have a strong disciplinary identity, regardless of the level (Mariotti et al., 2018). In a survey paper, Stylianides et al. (2024) highlight the central role of the content in many of the papers they selected and point out that among these papers, half of them concern the university level. More generally, the role of proof becomes increasingly important as one progresses through university studies, coherently with the potential role of proof and proving in conceptualization processes (e.g., Lakatos, 1976; Selden, 2012; Tall, 1992). This is developed in Sect. 4.2.

In recent years, the increasing and accelerated use of computational tools such as dynamic geometry systems, proof assistants, automated theorem provers, and, more generally, artificial intelligence-based tools has given rise to a great deal of research, including research on proof and proving (Hanna et al., 2019). We develop this in Sect. 4.3.

A third issue that we consider here (Sect. 4.4) concerns the assessment of students' proof productions and students' proof understanding by university teachers, which appears more challenging than expected (e.g., Conradie & Frith, 2000).

4.2 Contribution of argumentation and proof to mathematical understanding

As many researchers acknowledge, proof plays a crucial role in mathematics as a discipline and in the understanding and conceptualization of advanced students. This assumption is based on epistemological and didactic considerations documented in the literature (e.g., Hanna & De Villiers, 2012; Stylianides et al., 2024; Zaslavsky et al., 2012). The paper by Harel (2024, this issue) is part of a long-term study of what it means to understand mathematics and how to support the understanding of advanced mathematics students. The paper aims to present a conceptual framework called epistemological justification, which is located within the DNR-based instructional framework (Harel, 2008). The author views epistemological justification as a way of thinking, a form of mathematical understanding that revolves around the question: Why and how was a piece of mathematical knowledge [axiom, definition, theorem, counterexample, or abstract structure] conceived? Several illustrative examples in linear algebra are presented, taken from empirical studies. This paper is a solid contribution to the ongoing development of the DNR framework and opens avenues for further empirical studies.

A critical issue in the teaching and learning of proof is the relationship between argumentation and proof, which "holds great potential for helping teachers and students deal with the tension between the process leading up to the development of a student's proof and the requirements placed on the final product." (Durand-Guerrier et al., 2012, pp. 349–350). In the ZDM special issue *Mathematical Evidence and Argument*, published in 2019, a few papers deal with proof at the university level. Kempen and Biehler (2019) provide empirical evidence that pre-service teachers exposed to different types of proofs in multiple proof tasks and mathematical symbolic language in an innovative course entitled "Introduction to the Culture of Mathematics" begin to improve their proof competence. Aberdein (2019) "addresses from a philosophical perspective the relationships of evidence to proof, proof to derivation, argument to proof and argument to evidence, respectively," considering actual mathematical practices. He offers several unproven conjectures to illustrate how evidence and proof differ and argues with others that proof cannot be reduced to derivation. Finally, he argues that argumentation theory provides a common framework within which the three concepts can be subsumed. In a chapter of the book on practice-oriented research in tertiary mathematics education (Biehler et al., 2022), Kempen investigates the practice and meta-knowledge on proof and proving of 12 high school graduate students. He focuses on the basis for argumentation in elementary arithmetic; "the basis for argumentation consists of statements, definitions, and forms of reasoning that the student subjectively considers correct"

(Kempen, 2022, p. 206) and is expected to be common knowledge in school mathematics. The research shows that the individual basis for argumentation varied in terms of the mere elements and the epistemic value (certain, necessary, probable…) assigned to the statement involved in the tasks submitted. A primary result of the study is the role of degree of conviction, which can influence the individual difficulty level of a proving task and the construction of the proof. These results contribute to the ongoing discussion about the learning of proof in school mathematics.

Durand-Guerrier's (2024, this issue) is part of a broader project that consists of exploring the challenges students face in understanding completeness versus incompleteness, and the potential role of proofs and proving in fostering students' conceptualization (Durand-Guerrier & Tanguay, 2018). In this paper, the author presents an epistemological approach of R-completeness versus Q-incompleteness around the Intermediate Value Theorem and a multi-proofs activity around the Bolzano-Weierstrass Theorem. The goal is to make students aware that certain theorems they have previously learned in calculus or analysis need to be proved in an axiomatic system that includes a completeness axiom, in order to foster their understanding of the raison d'être and use of completeness axioms in proofs. This is the innovative feature of this study.

As mentioned above, the role of proof in the teaching and learning of advanced mathematics is crucial, so it is not surprising that proof-related issues are present in papers where proof is not the main focus. This is the case in four papers from various mathematical domains: graph theory (Sabra & Tabchi, 2024, this issue); symmetry (Hoffman & Biehler, 2024, this issue); linear algebra (Trigueros et al., 2024, this issue); and abstract algebra (Asada et al., 2024, this issue). These four papers contribute to shedding light on the intertwining of proof and proving and mathematical understanding in advanced university mathematics education.

4.3 The role of computer tools in proof and proving

Beyond the traditional role of logical issues in mathematics education (Dawkins & Roh, 2024; Durand-Guerrier et al., 2012; Epp, 2003) and the articulation between syntax and semantics (Alcock & Weber, 2004; Durand-Guerrier et al., 2012), new critical issues are appearing at the interface between mathematics and computer science (e.g., Avigad, 2019). The development of new tools, such as proof assistants, theorem provers, or tools based on generative artificial intelligence, provides digital assistance for discovery and proof (Borwein, 2012). Some mathematicians consider proof assistants relevant to research in mathematics (Bundy, 2011). They have recently raised educators' interest in their contribution to university mathematics education with two main facets: their contribution to the learning and teaching of logical knowledge and skills together with their contribution to improving proof and proving skills in mathematics (Avigad, 2019; Kerjean et al., 2022). Hanna et al. (2019), in a book published in Springer's *Mathematics Education in The Digital Area* series, emphasize the importance of initiating a dialogue between mathematics educators and researchers actively involved in developing such tools (e.g., Lodder, 2020). Not surprisingly, many of the book's chapters focus on the learning and teaching of proof and proving, including a few chapters on university mathematics education (e.g., Avigad, 2019; Komatsu & John, 2019).

Hanna et al., (2024, this issue) report on an exploratory study of integrating the proof assistant *Lean* into undergraduate teaching to support learning the mathematical proof concept. In the authors' view, the role of a proof assistant is to help undergraduates focus on the formal logical structure of proofs, an essential aspect of advanced proof and proving. They report on three undergraduate students' approach to proving with Lean the equivalence between a proposition *p* and its double negation *not (not p)*. The students involved in the experiment show creativity, flexibility, and autonomy. The diversity of the students' proofs is an original contribution to the state of the art, challenging the a priori idea that such tools might favor uniformity. This paves the way for further research on a larger scale involving proofs with specific mathematical content. This will also need to examine students' perceptions of such automatic provers (e.g. Iannone & Thoma, 2024).

Recently, the educational use of generative artificial intelligence has rapidly become a topic of interest to educators in general education and in various disciplines. With respect to mathematics. Yoon et al., (2024, this issue) argue for the use of ChatGPT in mathematical proof to help students construct and verify arguments, conjectures, and proofs. The authors have conducted clinical interviews with three undergraduate students who were encouraged to use ChatGPT in the proof tasks they were given. Analysis of the data highlights the role of three factors that influence students' use of ChatGPT: conceptions of mathematical proof, conceptions of generative artificial intelligence, and ethical considerations. The logical structure of the proof is also an aspect involved in checking the validity of the proofs produced by ChatGPT. This paper is a first contribution to the discussion of the relevance and pitfalls of using generative artificial intelligence such as ChatGPT to develop proof and logical skills in university mathematics education.

4.4 Assessment of proofs in mathematics

In advanced university mathematics, students are expected to be able to understand the proofs presented in lectures, handouts, or textbooks and to be able to construct their own proofs (Chartrand et al., 2018). However, research-based

evidence shows that understanding proofs remains challenging even for mathematics majors (e.g., Selden & Selden, 2003). According to Conradie and Frith (2000), a central challenge for researchers and educators is to find common ways of assessing students' proof comprehension beyond writing proofs identical or similar to those presented in class (Conradie & Frith, 2000). However, the results of a study by Inglis et al. (2013), who questioned 109 mathematicians about the validity of proofs in the context of a journal "supports the claims that there is not universal agreement among mathematicians regarding what constitutes a valid proof" (p. 279). The authors hypothesize that mathematicians as teachers may "be giving inconsistent feedback [to their students] on what features a proof needs to have to be considered as valid." (p. 280). In a recent study, Sommerhoff and Ufer (2019) analyzed and compared the acceptance criteria of school students, university students, and mathematicians in the context of teaching. Their results conclude university students' need for support in implementing acceptance criteria and suggest that a more explicit discussion of the functions of proofs, their evidential value, and the criteria for their acceptance may be beneficial. Such results plead for collaboration between mathematicians and educators for what concerns proof assessment in university mathematics education. More generally, the need for collaboration between mathematicians and mathematics educators is emphasized by Bardini et al. (2021), who present three case studies in Australia, the United States of America, and Mexico, or by Viirman (2018), who describes such collaboration in the case of a modeling project.

Cooley et al., (2024, this issue) describe a long-term project developed jointly by mathematicians and mathematics educators as a professional learning community. They have developed a qualitative framework, with the acronym PRIUM for *Promoting Reasoning in Undergraduate Mathematics*, whose main goal is to improve the assessment of proof comprehension. This framework can be used to develop assessment questions and to analyze students' responses, as well for individual, course, or program assessment, or can be implemented in a mathematics department. This paper shows the fruitfulness of developing long-term collaborations between mathematicians and mathematics educators, overcoming the institutional divide (Bardini et al., 2021), on a topic that is still under research at the university level: the assessment of students' understanding of proofs.

5 Discussion

We will summarize the key findings from the survey and the related papers of this special issue, following the three sections we used to structure our survey and the special issue.

5.1 Innovation

The section on innovations in higher mathematics education highlights the role of research-based perspectives on new ways to approach the development of participative advanced mathematics practices to foster students' learning. It addresses the role that novel teaching practices play in taking into account students' learning needs and technology's role in complementing and widening students' understanding of mathematics and its application. It also calls for attention to the role of motivation and other affective variables in students' advanced mathematics learning progress.

The first part of this section emphasizes the need for research-based approaches to improve student learning and overcome the continuing dominance of traditional lecturing methods. Significant learning advances have been made in the development of new theoretical approaches and methodologies based on theoretical frameworks that address the complexity of advanced mathematical concepts through active learning methods such as problem-based learning (PBL) and inquiry-based learning (IBL). Also, through the development of novel theory-based methodologies to carefully analyze topics that are known to be difficult for students, such as the model of meaning presented by Lankeit and Biehler (2024, this issue) in this issue, and the research possibilities that open up when a theoretical framework is challenged to solve new interesting phenomena related to student learning, as is the case in Villabona Millán et al., (2024, this issue).

Key findings highlight the importance of teacher expertise and ongoing professional development for teachers to effectively communicate complex ideas. Innovative strategies have been shown to promote deep learning, especially when integrated with technology. However, the variability in how different teachers implement these strategies underscores the need for professional development tailored to individual teaching styles and student needs. New ideas are needed to promote research on new approaches to proposing innovative, research-based instructional practices, as well as research on student and teacher learning outcomes when these novel proposals are used in the classroom.

Innovative practices, including capstone courses and student-centered activities, help bridge the gap between theoretical knowledge and practical application. Technology, such as computer algebra systems and dynamic geometry software, has enhanced student engagement and comprehension of complex concepts. The use of new technological devices such as artificial intelligence and programming needs attention to get a better idea of their possibilities and limitations when used in the classroom.

Future research should explore the impact of these strategies on student outcomes in diverse contexts. Investigating the long-term effects on understanding mathematical

concepts and examining how digital tools can enhance mathematical thinking are essential. Teacher development research, both on university teachers and future school teachers, is also critical to understanding.

Additionally, research on the motivational aspects of learning advanced mathematics is needed, focusing on how values, beliefs, and engagements influence success. Understanding these factors can inform interventions that improve academic performance and foster a deeper appreciation for mathematics.

5.2 Transition

We have identified the many transitions that occur from school to university and from university to the workplace, particularly for future mathematics teachers. In addition, transitions occur within universities, and we have focused on the transition between mathematics courses and courses in other disciplines where we find mathematics taught in context.

What transitions future mathematics teachers face and what kind of mathematics they need to positively influence their future teaching practice is a dynamic field of research. Among other things, a recent special issue of ZDM (see the survey paper by Wasserman et al., 2023) was devoted to this topic. The section on "Klein's double discontinuity and the transition from university to the mathematics teacher's workplace" identifies challenges in bridging university mathematics and secondary school teaching. This transition requires addressing affective and identity issues and adapting to different pedagogical practices. The three papers in this issue that address this topic suggest the design of new courses or new course activities for teachers that link university and school mathematics content processes and tasks in number theory, mapping geometry, and real numbers.

They highlight three different approaches that future course design and innovation might take. If secondary mathematics is to be taught in a more inquiry-based spirit, it is necessary to offer inquiry-based courses in university mathematics as well. However, students need to reflect on their own learning experiences in such courses as part of the course, and work on designing activities for future students based on these reflections and mathematical insights (Grenier-Boley and Robert 2024, this issue). Suppose that secondary mathematics is to be taught with an eye to university mathematics concepts. In this case, teachers need to learn to recognize the mathematical potential of students' questions and contributions and to experience that their acquired mathematical knowledge is helpful in such teaching situations, which need to be presented as model tasks in the mathematics course itself in order to influence teachers' self-efficacy with regard to the pedagogical potential of their mathematical knowledge (Hoffmann & Biehler, 2024, this issue). The topics discussed were symmetry and real numbers, but the approach could be applied to other mathematical topics. Empirical research on the effects of such courses on teachers' knowledge and beliefs is just beginning. In addition, design-based research involving systematic variation of course or activity design elements is still in its infancy. Nevertheless, there is a strong need to identify those components that produce the most promising results.

The section “Transitions between mathematics courses and courses in other university disciplines where mathematics is used” explores the challenges of aligning mathematics courses with applications in disciplines like engineering. Research focuses on understanding the gaps between these fields, often using Anthropological Theory of the Didactic (ATD) to analyze and design educational approaches. The one paper on this topic in this issue (Sabra & Tabchi, 2024, this issue) introduces an important new perspective to this area of research. Its key findings reveal variability in how teachers implement and use educational resources from the same pool. These findings highlight the need to consider the influence of individual teachers more generally than has been done in the field. The theoretical underpinnings of the "resource approach", whose relevance to engineering mathematics education was also highlighted in the review by Pepin et al. (2021), will provide a valuable foundation for this.

We have found different theoretical and practical approaches in working on the transitions between "university to school teaching" and "transitions between mathematics and other subjects within the university". However, we see a potential for cross-fertilization in theory and practice between these two fields of transition research that has not yet been noticed.

5.3 Proof and proving

In the section on "Proof and proving in advanced university mathematics," we highlight the critical role of proof in mathematical understanding, the new challenges raised by the integration of computer tools in proof and proving, and the challenges of proof assessment.

We have emphasized the role of proof in understanding advanced mathematical concepts in the literature review and in the articles included in this special issue where proof was the focus of the research or in articles where proof was an important secondary element. Several papers have emphasized the need for a fruitful collaboration between mathematicians and mathematics education researchers. This is consistent with the fact that in proof and proving, mathematical and logical issues are closely intertwined (Durand-Guerrier et al., 2012) so that it is not enough to master a general framework for dealing with proof in advanced university mathematics education. A deep understanding of the mathematics involved is required as the mathematical content

becomes more advanced (e.g., complex analysis, abstract algebra, topology, non-Euclidean geometry).

As in education in general, the interest in computer tools in mathematics is growing rapidly in the university mathematics education research community. The use of proof assistants in research in mathematics is one motivation for their introduction into university courses; another motivation is their potential contribution to the teaching and learning of logical skills, which are known to be challenging even at advanced levels. The development of proof assistants suitable for mathematics education is a dynamic interdisciplinary field at the interface between mathematics, computer science, and mathematics education. While their contribution to the development of logical skills is documented in the research literature, their contribution to the learning of proofs in advanced mathematical domains is still under research.

Recently, the introduction of generative artificial intelligence (AI), such as the human-like conversational ChatGPT, has raised new questions and concerns about its impact on students' autonomous work. Some mathematics education researchers are taking an innovative approach: Can such tools be used to help students develop proof skills? A general positive answer to this question is still pending. Another issue that needs to be addressed in the future is the long-term impact of the use of computer tools such as proof assistants and generative AI on logical and proof skills.

Assessment in the university teaching of mathematics in general, and proof in particular, remains a current area of research. The diversity of mathematical backgrounds of university teachers, which is common in mathematics departments, even in mathematics specialization programs, can lead to significant differences in teachers' assessment of their students' proofs and in their assessment of their students' understanding of proof. This is a challenge for mathematicians as teachers, for mathematics educators, for mathematics trainers, and for future mathematics teachers.

Although not addressed in this survey paper, proof and proving are also critical aspects of Klein's second discontinuity that require further investigation.

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